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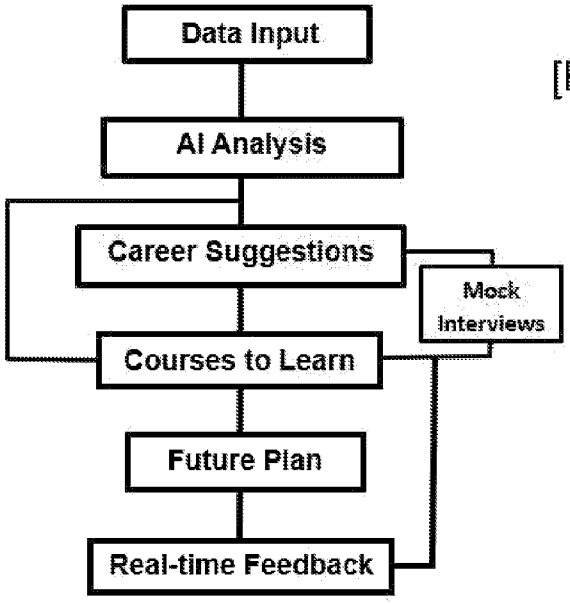
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(54) Title: AI-BASED CAREER COUNSELING SYSTEM WITH PREDICTIVE PLANS AND INDIVIDUAL GROWTH SUGGESTIONS



[Fig. 1]

(57) Abstract: The AI-Based Career Counseling System is a cutting-edge technology that will help in the process of career decision-making by using artificial intelligence (AI) and machine learning (ML). This system collects and processes real-time data on labor markets, user competencies, and career ambitions to offer tailored career advice, job-matching services, and skills development programs. Using predictive analytics and natural language processing (NLP), it predicts career trajectories, pinpoints emerging opportunities, and recommends specific programs of study. It is scalable across educational institutions, corporate HR departments, recruitment platforms, and government employment services to provide effective career guidance to individuals and enterprises. This invention disrupts traditional career counseling by automating the process and using data to get actionable insights.

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[Continued on next page]

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Description

Title of Invention: AI-Based Career Counseling System with Predictive Plans and Individual Growth Suggestions

[0001] The present invention relates generally to career counseling and more specifically to artificial intelligence and machine learning applications in career counseling. In this invention, we addressed data-driven decision-making systems that provide services, including personalized career advisory systems, skills assessment capabilities, job matching systems, predictive analytics of the career path, etc.

[0002] This system is a platform where AI and ML algorithms analyze users' skills, work history, interests, and market trends. It provides real-time job matching, tailored career advancement strategies, and skill-building suggestions. With the incorporation of artificial intelligence-powered evaluations and forecasting analytics, this system enhances career decision-making processes, allowing users to better and more efficiently design their professional paths.

Technical Field

[0003] G05D 101/10 – G06V 10/70 – G06Q 50/10

Background Art

[0004] 1- KR102590839

[0005] SYSTEM FOR PROVIDING CAREER COUNSELING CONTENT THROUGH NEURAL NETWORK-BASED CONTEXT ANALYSIS

[0006] Embodiments of the present invention provide a search word-based customized career design and recommendation system and a method thereof. The purpose of the present invention is to provide a system for providing a career counseling content through neural network-based context analysis. According to the embodiment of the present invention, the method thereof includes: a step of receiving multiple pieces of user information about each of multiple users from an outer server, wherein multiple pieces of user information include: information corresponding to at least one among age, sex, score by each subject, and learning ability by each subject, about each user; and first user information; a step of receiving a first career aptitude test result from a first user terminal; a step of determining an upper occupational cluster corresponding to a first user among the multiple upper occupational clusters based on the first career aptitude test result, wherein the number of the multiple upper occupational clusters is 10; a step of realizing a lower occupational cluster classification model about each of the multiple upper occupational clusters; a step of receiving a first input text from the first user terminal; and a step of determining a lower occupational cluster corresponding to the first user based on the first input text.

[0007] The mentioned patent is close to our claimed one but it is not real-time nor based on Artificial intelligence, it also does not predict the future path of the users and suggest to them their required skills.

[0008] 2- CN102630322

[0009] Career-counseling system and method

[0010] The present invention relates to a career-counseling method for a career-counseling system comprising a server which can be connected to a plurality of subscriber terminals via a network. The career-counseling method comprises: a step of receiving information from a subscriber terminal; a step of providing the subscriber terminal with a plurality of psychological tests; a step of receiving answers to the psychological tests from the subscriber terminal; a step of analyzing the type of psychology on the basis of the received answers; a step of extracting a plurality of career groups on the basis of the type of psychology analyzed in the previous step; a step of dividing paths for reaching each of the extracted career groups into stages; a step of preparing a report including content on the behavior required in each stage; and a step of transmitting the report to the subscriber terminal.

[0011] The mentioned invention is similar to our claimed one but it only has some of our system's features such as counseling by tests and analyzing them, while in the claimed system some beneficial courses and certificates for users are also available for their personal growth.

[0012] 3- IN202241014042

[0013] AI enabled Online Interview/Counselling/Pitching software platform

[0014] The present invention provides a System, method and Software for individuals to experience an interview Simulation and develop career, interview and pitching skills. It allows individuals to experience a full interview Simulation, including pre-andpost-interview Stages. The invention allows individuals to communicate with a computer generated interviewer character. It simulates a discussion by Speaking to the individual and asking the individual job-related questions, and displaysoutput on the computer terminal and/or digitizes statements into Speech. The individual responds to the Statements by typing replies and/or speaking replies into a device Such as a microphone, Video camera or telephony device that receives and records the responses onto the System. Once the interview is complete, the individual can review all his/her responses via a customized computer interface. The invention allows organizations to Screen potential employees by conducting initial Screening interviews. It allows individuals to self-screen by seeing which jobs they would be interested in and by Submitting pre-Screened data to employers. Finally, it allows individuals to train for interviews by going on realistic practice job interviews. The invention is able to

provide detailed analysis and recommendations regarding the practice interviews to users, which assists them in developing career and interview skills.

[0015] The said patent is similar to the claimed career counseling system, but it mostly focuses on tests and training for interviews based on artificial intelligence and does not offer real job offers, while our claimed patent contains all of these features.

[0016] 4- US20190325542

[0017] Survey submission system and method for personalized career counseling

[0018] A survey submission system and method for a personalised career counselling is provided to multiple users. The system includes one or more display devices with a display unit, a display region variation determining unit, a preloading controller, and a preloading performing unit displaying multiple program parameters in proximity to a short text field. Particularly, one or more display devices includes multiple display device modules and display device modules is selected from any one of an user interface module, an admin interface module, a sponsor interface module admin interface module and an expert interface module. Moreover, the system also has a remote server for storing a multiple of databases for storing multiple data related to program parameters. Particularly, one or more display device and the remote server communicate via a communication channel enabling users to communicate with other users and experts in a highly secured chat room. Moreover, communication is via secure chat room enabling users to share information only with user consent. The system analyses the answers by a trend data analysis.

[0019] This mentioned system operates for each individual and their career, but it only refers to one of the features that our claimed platform has, especially since it gets feedback from the users and gives feedback to them in real time.

[0020] 5- United States Patent 12198213

[0021] Systems and methods for personality analysis for selection of courses, colleges, and/or careers

[0022] Methods, computer-readable media, software, and career planning system may receive inputs from a user in response to questions related to personality and may provide outputs related to preferences for schooling and other career options, such as courses, colleges, universities, programs, majors, careers, jobs, and/or companies. The personality of the user may be broken into four (or other numbers) different pillars, life vectors, and/or personality characteristics and the user may be provided a score in each of the pillars, life vectors, and/or personality characteristics based on the responses to the questions. These scores for each of the pillars, life vectors, and/or personality characteristics may be then matched to various schooling and career option recommendations for the user.

[0023] The said patent is similar to our claimed one especially because it plans for the user's career but there are no mock interviews or career improvement items mentioned in it.

Summary of Invention

[0024] A career counseling system uses artificial intelligence, machine learning, and natural language processing to suggest job positions and create a personal plan. It collects users' educational and personal data for AI analysis of strengths, weaknesses, and career paths to form a profile. The system updates recommendations based on market trends and user input, predicts successful career options, and suggests relevant courses for their future. Users can also engage in mock interviews with feedback.

Technical Problem

[0025] Traditional career counseling does not offer personalized insights and cannot adapt to real-time labor market dynamics. Moreover, predictive career analytics are missing, so long-term career planning leads to ineffectiveness in acquiring skills and seeking jobs. The proposed invention overcomes these limitations by employing AI-driven automation as well as predictive modeling.

[0026] As mentioned, current career counseling systems typically ignore real-time adaptability, personalized recommendations, and predictive insights. However, AI-driven approaches for more complex career counseling have opened up to help individuals with dynamic and customized career planning.

Solution of Problem

[0027] The AI-based career counseling system consists of the following features:

[0028] - Data Collection: Sourcing and analyzing real-time labor market data, job descriptions, industry trends, and educational materials.

[0029] - Machine Learning Algorithms: Helping identify potential career paths, appropriate skillsets, and relevance to possible career opportunities through supervised and unsupervised learning.

[0030] - Personalized Career Guidance: Providing customized career advice by taking into account a person's skills, experience, and market requirements.

[0031] - Career trajectory predictions based on historical data and trend analysis.

[0032] - System Architecture Diagram: Depicts data flow between users, AI modules, job databases, and educational resources.

[0033] - Potential UI Mockup: a mockup of a sample dashboard with personalized job recommendations and skill development insights.

[0034] - Predictive Career Path Model with AI: Shows the career path prediction based on user input data.

[0035] - ML Workflow: a pipeline showing how the data flows from input to model training to recommendation generations to updates in real-time.

[0036] How the system runs:

[0037] - User Profile Analysis:

[0038] The system gathers user data, including education, work experience, skill sets, and career aspirations. Then AI algorithms analyze strengths, weaknesses, and future career paths.

[0039] - Intelligent Job Matching:

[0040] NLP models process job descriptions and user profiles to recommend the most suitable career opportunities. We update suggestions following market trends and user interaction.

[0041] - Predictive Career Analytics:

[0042] It uses statistical models to predict career options that would get you higher degree outcomes versus the degree you achieved. This gives users forecast insights into potential earnings, career growth, and industry demand.

[0043] - Skill Development Framework:

[0044] The system also suggests relevant courses, certifications, and training programs according to each person's career objectives. By providing continuous feedback on how users progress, recommendations are further refined.

[0045] - Interview Preparation using AI:

[0046] Automated mock interview modules using NLP sentiment analysis and scoring algorithms to evaluate user responses. Users also get real-time feedback to improve their performance in interviews.

Advantage Effects of the Invention

[0047] - Improved Personalization: Equinor aims for AI-powered recommendations to suit personal goals, strengths, and labor market trends.

[0048] - Efficient Job Matching Advanced NLP and deep learning algorithms for accurate job matches.

[0049] - Data-Driven Insights: Algorithms analyze vast amounts of career and job market data, highlighting in-demand skills.

[0050] - Dynamic Skill and Career Development Plans: Users get workflows on how to improve and advance their careers automatically.

[0051] - Scalability: The solution is industry-agnostic and can be deployed in educational institutes, increasing the reach and efficiency of career counseling.

Brief Description of Drawings

[0052] [Fig.1] Shows a flowchart that depicts the general steps taken in the system.

Description of Embodiments

[0053] [Fig.1] Presents the progress of the career counseling system from when a user enters data and AI modules analyze data, then based on the user's personality and

background, the NLP modules suggest jobs, and the system offers some courses and skills for users to learn and obtain certificates to help them improve their abilities, the users will be tested and their results will be given real-time, also the users can report their feedback, and overall, they will have their upcoming plan for their career.

Examples

[0054] There are some detailed examples of how the claimed system works:

[0055] 1- From a marketing background with social media experience, where the model understood that a transition towards digital marketing certifications would assist in transitioning into a more data-driven role.

[0056] 2- An IT professional looking for an AI/ML role is recommended to upskilling programs on AI/ML or job opportunities based on their coding proficiency.

[0057] 3- A mid-career finance professional is coached in changing careers into a fintech role, given data on market trends and which roles to target

Industrial Applicability

[0058] The claimed system can be used in the following fields of different industries:

[0059] - Educational Institutions: Improved career counseling services for students and graduates.

[0060] - Corporate HR departments: Supporting employees in career development and internal mobility.

[0061] - Job Search Platforms: Making it easier for seekers and employers to find and connect.

[0062] - Government Employment Services: Providing the data-driven insights you need to develop a workforce and plan employment policy.

[0063] With the amalgamation of AI and ML, this system revolutionizes career counseling by converting it into a dynamic and data-driven process on the whole, making accurate, time-saving, and efficient career advisory services for individuals and organizations.

Claims

[Claim 1] A career counseling system based on artificial intelligence, machine learning, and natural language processing to recommend job positions and predictive analytics with a real-time personal plan.

[Claim 2] According to claim 1, the system gathers users' educational and personal data, so the AI algorithms analyze strengths, weaknesses, and future career paths to generate a personal profile.

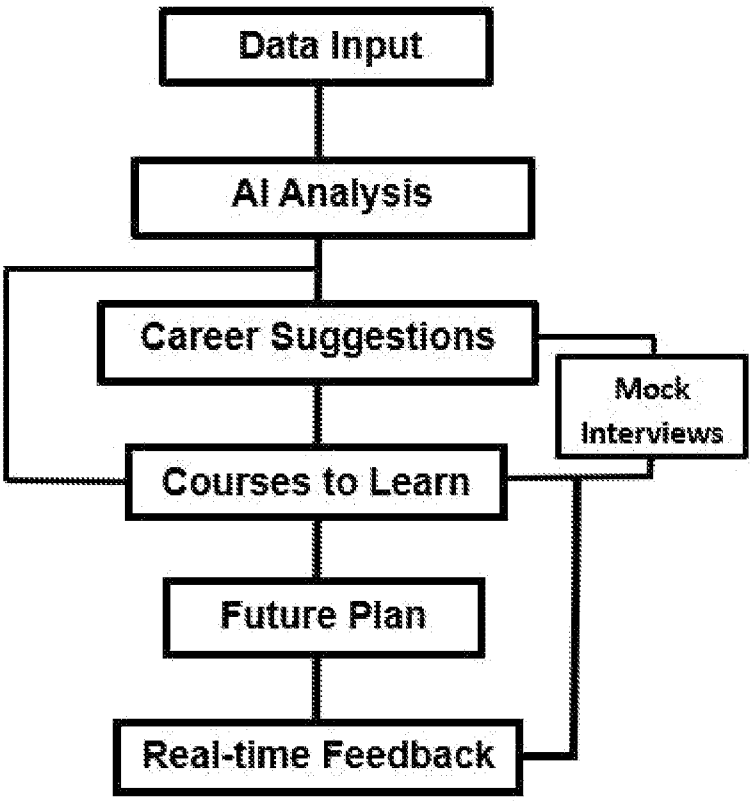
[Claim 3] According to claim 2, the NLP models process job descriptions and other user profiles to recommend the appropriate career occasions while the system updates the suggestions following market trends and user interaction.

[Claim 4] According to claim 3, The system practices statistical models to predict career options that would get the user a higher degree of outcomes which increases the users' forecast insights into potential earnings, career growth, and general demands.

[Claim 5] According to claim 4, relevant courses and training programs will be introduced to the users according to each person's career objectives, and by continuous feedback, they will be upgraded.

[Claim 6] According to claim 5, users can have automated UI mock interview modules with NLP sentiment analysis and scoring algorithms to estimate their outcomes in real-time feedback through the ML workflow.

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[Fig. 1]

INTERNATIONAL SEARCH REPORT

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A. CLASSIFICATION OF SUBJECT MATTER
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According to International Patent Classification (IPC) or to both national classification and IPC

B. FIELDS SEARCHED

Minimum documentation searched (classification system followed by classification symbols)

G06Q, G06N, G06F

Documentation searched other than minimum documentation to the extent that such documents are included in the fields searched

Electronic database consulted during the international search (name of database and, where practicable, search terms used)

PatSeer, IPO Internal Database

C. DOCUMENTS CONSIDERED TO BE RELEVANT

Category*	Citation of document, with indication, where appropriate, of the relevant passages	Relevant to claim No.
X	WO2018026961A1 (PAYSA, INC.) 08 FEBRUARY 2018 (08-02-2018) [ABSTRACT, CLAIMS 1-13, PARAGRAPHS 23-27, 43-47, 115-117, 141-143]	1-6

☐ Further documents are listed in the continuation of Box C. ☒ See patent family annex.

* Special categories of cited documents:
"A" document defining the general state of the art which is not considered to be of particular relevance
"D" document cited by the applicant in the international application
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"Y" document of particular relevance; the claimed invention cannot be considered to involve an inventive step when the document is combined with one or more other such documents, such combination being obvious to a person skilled in the art
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INTERNATIONAL SEARCH REPORT
Information on patent family members

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WO 2018026961 A1	08-02-2018	US 2018039946 A1	08-02-2018